1 Smooth Vector Fields

1.1 Vector Fields on Manifold

Definition 1.1.1

If M is a smooth manifold with or without boundary, a ***vector field on*** M is a section of the map $\pi : TM \to M$. More concretely, a vector field is a continuous map $X : M \to TM$, usually written $p \mapsto X_p$, with the property that

$$\pi \circ X = \mathrm{Id}_M, \tag{1.1}$$

or equivalently, $X_p \in T_pM$ for each $p \in M$.

Remark.

- For a "vector field" that may be not contiuous, we call it ***rough vector field on*** M .
- Under the canonical smooth structure of M and TM , we can define the smoothness of X.

Definition 1.1.2 ▶ component functions of a vector field

Suppose M is a smooth n-manifold (with or without boundary). If $X : M \to TM$ is a rough vector field and $(U, (x^i))$ is any smooth coordinate chart for M, we can write the value of X at any point $p \in U$ in terms of the coordinate basis vectors:

$$X_p = X^i(p) \frac{\partial}{\partial x^i}\bigg|_p . \tag{1.2}$$

This defines n functions $X^i : U \to \mathbb{R}$, called the ***component functions of*** X in the given chart.

Proposition 1.1.3 ▶ Smoothness Criterion for Vector Fields

Let M be a smooth manifold with or without boundary, and let $X : M \to TM$ be a rough vector field. If $(U, (x^i))$ is any smooth coordinate chart on M, then the restriction of X to U is smooth if and only if its component functions with respect to this chart are smooth.

Proposition 1.1.4

Let M be a smooth manifold with or without boundary, and let $X : M \to TM$ be a rough vector field. The following are equivalent:

(a) X is smooth.

(b) For every $f \in C^\infty(M)$, the function Xf is smooth on M.

(c) For every open subset $U \subseteq M$ and every $f \in C^\infty(U)$, the function Xf is smooth on U.

Definition 1.1.5 ▶ Vector Field as a Derivation

A smooth vector field $X \in \mathfrak{X}(M)$ defines a map

$$\begin{aligned} X : C^\infty(M) &\to C^\infty(M) \\ f &\mapsto Xf \end{aligned}$$

which is linear over $\mathbb{R}$ and following the product rule for vector fields

$$X(fg) = fXg + gXf$$

We call this map a ***Derivation on*** $C^\infty(M)$.

Proposition 1.1.6

Let M be a smooth manifold with or without boundary. A map $D : C^\infty(M) \to C^\infty(M)$ is a derivation if and only if it is of the form $Df = Xf$ for some smooth vector field $X \in \mathfrak{X}(M)$.

Proposition 1.1.7

Suppose M and N are smooth manifolds with or without boundary, and $F : M \to N$ is a diffeomorphism. For every $X \in \mathfrak{X}(M)$, there is a unique smooth vector field on N that is F-related to X.

1.2 Vector Field and Smooth Maps

Definition 1.2.1 ▸ F -related vector fields

Suppose $F : M \to N$ is smooth and X is a vector field on M, and suppose there happens to be a vector field Y on N with the property that for each $p \in M$, $dF_p(X_p) = Y_{F(p)}$. In this case, we say the vector fields X and Y are F-***related***.

Remark.
这是 F-相关向量场的逐点定义.

Proposition 1.2.2

Suppose $F : M \to N$ is a smooth map between manifolds with or without boundary, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Then X and Y are F-related if and only if for every smooth real-vealued function f defined on an open subset of N,

$$X(f \circ F) = (Yf) \circ F$$

Remark.

- 这是 F-相关向量场的 $C^\infty_{F(p)}(M)$-导子刻画.
- F 参与了以下活动: 将函数芽 $f \in C^\infty_{F(p)}(N)$ 的定义域拉回到 M 上以便于 X 作用, 以及将 (Yf) 的定义域 (Y 的取值) 拉回到 M 上以便于比较.

Proof. For any $p \in M$ and any $f \in C^\infty_{F(p)}(M)$,

$$X(f \circ F)(p) = X_p(f \circ F) = \mathrm{d}F_p(X_p) f$$

while

$$(Yf) \circ F(p) = (Yf)(F(p)) = Y_{F(p)} f$$

□